

10th grade - Learning evidence 2.1: Analysis of Decisions

Name: _____

Assessment and evaluation

This exam evaluates only the criteria C1–C5. Each criterion is evaluated across the three points below. A student passes a criterion if it is correctly activated in at least two of the three points. The overall exam result is binary (Passed/Not passed) and is determined by the set of criteria passed.

1. A school cafeteria must choose one of three menu plans for a two-week festival: Plan A (local produce focus), Plan B (mixed menu), or Plan C (bulk staples). Demand for meals can be high, medium, or low. Based on past festivals, the probability of high demand is 0,35, medium demand is 0,45, and low demand is 0,20. Profits are measured in hundreds of dollars. If demand is high, the profits are 92 for Plan A, 70 for Plan B, and 55 for Plan C. If demand is medium, the profits are 50 for Plan A, 62 for Plan B, and 48 for Plan C. If demand is low, the profits are –8 for Plan A, 30 for Plan B, and 38 for Plan C.
 - Formulate the decision problem by clearly stating the decision alternatives, states of nature, probabilities, and consequences.
 - Interpret the key elements of the decision problem (alternatives, fortuitous events, and consequences) in this context.
 - Build the payoff table from the description.
 - Apply the maximax rule and state the recommended plan.
 - Apply the maximin rule and state the recommended plan.
2. A delivery cooperative is choosing a fleet strategy for the next season: Strategy A (electric vans), Strategy B (hybrid vans), or Strategy C (diesel vans). Demand for deliveries can be strong, steady, or weak. Fuel costs can be low or high. The cooperative estimates the demand probabilities as 0,30 (strong), 0,50 (steady), 0,20 (weak). Fuel costs are independent of demand, with probabilities 0,55 (low) and 0,45 (high).

Revenue (in thousands of dollars) by demand level is shown below.

Demand level	Strategy A	Strategy B	Strategy C
Strong	420	380	340
Steady	320	300	270
Weak	210	230	220

Operating costs (in thousands of dollars) by fuel cost level are shown below.

Fuel cost level	Strategy A	Strategy B	Strategy C
Low	160	190	210
High	220	250	290

- Formulate the decision problem and define the combined states of nature using the independence assumption.
 - Interpret the decision alternatives, states of nature, and consequences for this setting.
 - Build the payoff table by computing profits for each combined state and the probability of each state.
 - Apply the maximax rule and state the recommended strategy.
 - Apply the maximin rule and state the recommended strategy.
3. A community sports center must choose a membership model for the next year: Model A (standard monthly pass), Model B (premium pass with classes), or Model C (flex pass). Demand for memberships can be high, medium, or low. Staffing costs can be favorable or unfavorable. Demand probabilities are 0,25 (high), 0,50 (medium), 0,25 (low). Staffing costs are independent of demand, with probabilities 0,60 (favorable) and 0,40 (unfavorable).

Expected number of members by demand level is shown below.

Demand level	Model A	Model B	Model C
High	620	480	700
Medium	460	360	520
Low	300	240	360

Membership fees are 35 (Model A), 48 (Model B), and 30 (Model C) dollars per member per year. Variable staffing cost per member is 18 when costs are favorable and 26 when costs are unfavorable. Fixed annual costs are 6,500 for Model A, 9,000 for Model B, and 5,200 for Model C. All monetary amounts in this point can be treated in dollars.

- Formulate the decision problem, specifying how states of nature are formed and how profits are computed.
- Interpret the alternatives, fortuitous events, and consequences in this context.
- Build the payoff table by calculating profit for each combined state and assigning the correct probabilities.
- Apply the maximax rule and state the recommended model.
- Apply the maximin rule and state the recommended model.